Model metrics (estimate [95% bootstrap CI])

metric	RF_Luke	XGB_Thomas	Logit_Morgan
brier	0.1021 [0.0963, 0.1078]	0.1014 [0.0955, 0.1072]	0.1225 [0.1156, 0.1294]
log_loss	0.3158 [0.2992, 0.3323]	0.3109 [0.2943, 0.3274]	0.3804 [0.3601, 0.4001]
brier_skill	0.5609 [0.5279, 0.5945]	0.5637 [0.5301, 0.5970]	0.4732 [0.4357, 0.5132]
roc_auc	0.9287 [0.9189, 0.9377]	0.9304 [0.9210, 0.9391]	0.8932 [0.8791, 0.9068]
average_precision	0.8991 [0.8749, 0.9192]	0.9000 [0.8767, 0.9203]	0.8594 [0.8273, 0.8874]
max_youden_ks	0.6907 [0.6705, 0.7159]	0.6935 [0.6706, 0.7166]	0.6105 [0.5845, 0.6448]
cal_intercept	-0.0601 [-0.1330, 0.0142]	0.0107 [-0.0690, 0.0909]	-0.0318 [-0.1102, 0.0440]
cal_slope	1.0526 [0.9942, 1.1139]	0.9781 [0.9289, 1.0296]	0.9055 [0.8530, 0.9600]
ece	0.0101 [0.0092, 0.0205]	0.0109 [0.0082, 0.0203]	0.0123 [0.0102, 0.0240]
accuracy	0.8511 [0.8407, 0.8611]	0.8511 [0.8410, 0.8611]	0.8227 [0.8107, 0.8343]
balanced_accuracy	0.8322 [0.8171, 0.8471]	0.8316 [0.8162, 0.8467]	0.7943 [0.7751, 0.8128]
tpr_recall	0.7611 [0.7320, 0.7899]	0.7578 [0.7290, 0.7870]	0.6871 [0.6500, 0.7218]
tnr_specificity	0.9034 [0.8925, 0.9133]	0.9054 [0.8945, 0.9156]	0.9015 [0.8914, 0.9110]
ppv_precision	0.8207 [0.7929, 0.8456]	0.8232 [0.7964, 0.8485]	0.8022 [0.7696, 0.8303]
npv	0.8668 [0.8569, 0.8760]	0.8654 [0.8559, 0.8750]	0.8322 [0.8209, 0.8429]
f1	0.7898 [0.7638, 0.8147]	0.7891 [0.7629, 0.8140]	0.7402 [0.7073, 0.7700]
mcc	0.6759 [0.6469, 0.7037]	0.6758 [0.6465, 0.7043]	0.6110 [0.5753, 0.6440]
cohen_kappa	0.6747 [0.6458, 0.7026]	0.6744 [0.6452, 0.7029]	0.6068 [0.5714, 0.6403]
youden_j	0.6645 [0.6342, 0.6943]	0.6632 [0.6323, 0.6935]	0.5886 [0.5503, 0.6255]
expected_cost	0.1489 [0.1389, 0.1593]	0.1489 [0.1389, 0.1590]	0.1773 [0.1657, 0.1893]
cost_vs_trivial	0.4052 [0.3582, 0.4547]	0.4050 [0.3582, 0.4532]	0.4824 [0.4291, 0.5401]
brier_MCB	0.0015	0.0014	0.0018
brier_DSC	0.1319	0.1324	0.1118
brier_UNC	0.2325	0.2325	0.2325
logloss_MCB	0.0053	0.0050	0.0089
logloss_DSC	0.3471	0.3517	0.2862
spiegelhalter_z	-3.4644	2.3857	4.5265
spiegelhalter_p	<0.001	0.017	<0.001
n_extreme_probs	0	0	0
n_confident_misses	0	0	0
roc_auc_delong_se	0.0019	0.0018	0.0025

Paired comparisons (difference = first model minus second)

RF_Luke minus XGB_Thomas

metric	difference [95% CI]	test	p	p (Holm)
brier	0.00067 [-0.00053, 0.00191]	Diebold-Mariano	0.090	0.090
log_loss	0.00490 [0.00153, 0.00836]	Diebold-Mariano	<0.001	<0.001
roc_auc	-0.00169 [-0.00359, 0.00008]	DeLong	0.002	0.002
average_precision	-0.00090 [-0.00347, 0.00161]	paired bootstrap	0.458	0.458
accuracy	-0.00006 [-0.00424, 0.00408]	McNemar mid-p	0.969	0.969
balanced_accuracy	0.00066 [-0.00406, 0.00497]	paired bootstrap	0.822	0.822
f1	0.00067 [-0.00542, 0.00637]	paired bootstrap	0.874	0.874
mcc	0.00012 [-0.00904, 0.00919]	paired bootstrap	0.972	0.972
expected_cost	0.00006 [-0.00408, 0.00424]	paired bootstrap	0.942	0.942
cal_slope	0.07451 [0.05007, 0.09936]	paired bootstrap	<0.001	<0.001
ece	-0.00079 [-0.00735, 0.00909]	paired bootstrap	0.834	1.000

RF_Luke minus Logit_Morgan

metric	difference [95% CI]	test	p	p (Holm)
brier	-0.02037 [-0.02484, -0.01593]	Diebold-Mariano	<0.001	<0.001
log_loss	-0.06457 [-0.07690, -0.05278]	Diebold-Mariano	<0.001	<0.001
roc_auc	0.03545 [0.02810, 0.04370]	DeLong	<0.001	<0.001
average_precision	0.03967 [0.02925, 0.05224]	paired bootstrap	<0.001	<0.001
accuracy	0.02838 [0.01815, 0.03851]	McNemar mid-p	<0.001	<0.001
balanced_accuracy	0.03793 [0.02770, 0.04859]	paired bootstrap	<0.001	<0.001
f1	0.04960 [0.03532, 0.06502]	paired bootstrap	<0.001	<0.001
mcc	0.06486 [0.04235, 0.08774]	paired bootstrap	<0.001	<0.001
expected_cost	-0.02838 [-0.03851, -0.01815]	paired bootstrap	<0.001	<0.001
cal_slope	0.14710 [0.07928, 0.21414]	paired bootstrap	<0.001	<0.001
ece	-0.00225 [-0.01198, 0.00637]	paired bootstrap	0.694	1.000

XGB_Thomas minus Logit_Morgan

metric	difference [95% CI]	test	p	p (Holm)
brier	-0.02104 [-0.02529, -0.01662]	Diebold-Mariano	<0.001	<0.001
log_loss	-0.06947 [-0.08115, -0.05823]	Diebold-Mariano	<0.001	<0.001
roc_auc	0.03714 [0.02995, 0.04525]	DeLong	<0.001	<0.001
average_precision	0.04057 [0.03026, 0.05315]	paired bootstrap	<0.001	<0.001
accuracy	0.02843 [0.01784, 0.03881]	McNemar mid-p	<0.001	<0.001
balanced_accuracy	0.03728 [0.02694, 0.04811]	paired bootstrap	<0.001	<0.001
f1	0.04893 [0.03437, 0.06473]	paired bootstrap	<0.001	<0.001
mcc	0.06474 [0.04207, 0.08832]	paired bootstrap	<0.001	<0.001
expected_cost	-0.02843 [-0.03881, -0.01784]	paired bootstrap	<0.001	<0.001
cal_slope	0.07258 [0.01260, 0.13200]	paired bootstrap	0.014	0.014
ece	-0.00146 [-0.01060, 0.00507]	paired bootstrap	0.485	1.000

Per-group metrics (by hotel)

Resort Hotel (n = 5,785, prevalence 0.269)

metric	RF_Luke	XGB_Thomas	Logit_Morgan
brier	0.0969 [0.0915, 0.1030]	0.0949 [0.0893, 0.1010]	0.1235 [0.1160, 0.1315]
log_loss	0.3071 [0.2913, 0.3275]	0.2983 [0.2818, 0.3179]	0.3844 [0.3633, 0.4068]
roc_auc	0.9233 [0.9122, 0.9340]	0.9262 [0.9156, 0.9364]	0.8720 [0.8564, 0.8864]
cal_intercept	-0.0333 [-0.1259, 0.0550]	0.0436 [-0.0502, 0.1385]	0.0267 [-0.0855, 0.1305]
cal_slope	1.1355 [1.0421, 1.2345]	1.0507 [0.9772, 1.1321]	0.9221 [0.8622, 0.9897]
accuracy	0.8598 [0.8484, 0.8706]	0.8598 [0.8480, 0.8696]	0.8154 [0.7996, 0.8289]
mcc	0.6352 [0.6002, 0.6705]	0.6339 [0.5994, 0.6665]	0.5009 [0.4525, 0.5441]
expected_cost	0.1402 [0.1294, 0.1516]	0.1402 [0.1304, 0.1520]	0.1846 [0.1711, 0.2004]

City Hotel (n = 12,118, prevalence 0.415)

metric	RF_Luke	XGB_Thomas	Logit_Morgan
brier	0.1046 [0.0970, 0.1124]	0.1045 [0.0969, 0.1121]	0.1220 [0.1132, 0.1309]
log_loss	0.3199 [0.2982, 0.3420]	0.3169 [0.2955, 0.3380]	0.3784 [0.3522, 0.4056]
roc_auc	0.9284 [0.9157, 0.9392]	0.9295 [0.9177, 0.9402]	0.8986 [0.8801, 0.9143]
cal_intercept	-0.0730 [-0.1695, 0.0270]	-0.0047 [-0.0977, 0.0991]	-0.0598 [-0.1618, 0.0420]
cal_slope	1.0106 [0.9343, 1.0874]	0.9413 [0.8811, 1.0024]	0.8999 [0.8307, 0.9682]
accuracy	0.8469 [0.8347, 0.8603]	0.8470 [0.8342, 0.8598]	0.8262 [0.8110, 0.8415]
mcc	0.6826 [0.6484, 0.7158]	0.6828 [0.6440, 0.7149]	0.6391 [0.5973, 0.6762]
expected_cost	0.1531 [0.1397, 0.1653]	0.1530 [0.1402, 0.1658]	0.1738 [0.1585, 0.1890]